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RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

Section 9

Explosive Atmosphere

Important Notice

Information contained in these sections is pertinent to all test procedures described in the other sections of this document. Further, Appendix A is applicable for identifying environmental tests performed.

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9.0 Explosive Atmosphere

9.1 Purpose of the Test

This test specifies requirements and procedures for aircraft equipment that may come into contact with flammable fluids and vapors such as those specified herein. It also refers to normal and fault conditions that could occur in areas that are or may be subjected to flammable fluids and vapors during flight operations.

The flammable test fluids, vapors or gases referred to in this section simulate those normally used in conventional aircraft and that require oxygen for combustion (e.g., monofuels are not included).

These standards do not relate to potentially dangerous environments occurring as a result of leakage from goods carried on the aircraft as baggage or cargo.

Note: The explosive atmosphere tests shall not be conducted prior to any other DO160/ED-14 tests except flammability test (see Subsection 3.2, "Order of Tests").

9.2 Explosion Proof

Equipment is explosion proof when it has been determined that there is negligible risk that it will cause an explosion of a flammable gas or vapor within the declared environment.

9.3 Equipment Categories

9.3.1 Category A Equipment

Category A equipment is designed so that:

a. In normal operation the temperature of any surface will not rise to a level capable of causing ignition, nor will any operating part cause ignition so that it meets the non ignition test specified in paragraph 9.6.2.

b. Ignition of an explosive mixture is contained within the equipment without igniting an explosive atmosphere surrounding it so that it meets the Containment tests specified in paragraph 9.6.1.

Hermetically sealed equipment meeting subparagraph 9.3.1 b. shall be identified as Category A equipment.

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9.3.2 Category E Equipment

Category E equipment is not hermetically sealed and not contained in cases designed to prevent flame and explosion propagation.

Such equipment is designed so that in operation the temperature of any internal or external surface will not rise to a level capable of causing ignition, nor will any operating internal part cause ignition so that it meets the non ignition test specified in paragraph 9.6.2.

9.3.3 Category H Equipment

Category H equipment contains hot spot surfaces (external or internal) and are non-spark producing under normal operating conditions (see paragraph 4.6.1).

Such equipment designed so that in normal operation the temperature of any internal or external surfaces will not rise to a level capable of causing ignition so that it meets the components or surface temperature test of paragraph 9.6.3.

Hermitically sealed equipment meeting temperature requirement on external surface shall be identified as Category H equipment.

9.4 General Test Requirements

9.4.1 General

The test requirements specified below are necessary to assure that the equipment, when tested in accordance with the applicable test procedure, will comply with the standards established in Subsection 9.3.

9.4.2 Test Specimen

The test specimens selected shall be representative of production equipment.

9.4.3 Fuel

Unless otherwise specified, the fuel used may be propane, normal hexane and associated equations.

9.4.4 Fuel Mixtures

a. For propane, a 1.05 stoichiometric mixture of 3.85% to 4.25% by volume of propane to 96.15% to 95.75% by volume of air. For an example of test apparatus see Figure 9-1.

b. For hexane, a 1.80 stoichiometric fraction of normal hexane shall be calculated according to the following equations:

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(1) Volume (ml) of 95% normal hexane (metric units) =

$$(4.27 \times 10^{-4}) \frac{[\text{net chamber vol (liters)}][\text{chamber pressure (pascals)}]}{[\text{Chamber temp (K)}][\text{Relative density of n-hexane}]}$$

(2) Volume (ml) of 95% normal hexane (English units) =

$$(150.41) \frac{[\text{net chamber vol (ft}^3\text{)}][\text{chamber pressure (psia)}]}{[\text{Chamber temp (R)}][\text{Relative density of n-hexane}]}$$

Note: K = the thermodynamic temperature and is °C + 273.15
R = the thermodynamic temperature and is °F + 459.67
The relative density of normal hexane can be determined from Figure 9-2.

The equipment used to vaporize the fuel for use in the explosive atmosphere test should be so designed that a small quantity of air and fuel vapor will be heated together to a temperature such that the fuel vapor will not condense as it is drawn from the vaporizer into the chamber.

When the test facility is designed for fuel vaporization inside the explosion chamber, the fuel may be introduced at the ambient temperature of the test site.

9.5 Equipment Design and Installation Information

Equipment specifications should detail any design constraints applicable to the particular category of equipment enclosure. Such design constraints should include the following as appropriate:

a. Equipment that may come into contact with flammable fluids or vapors and that in normal operations may produce arcs, sparks or hot surfaces shall be designed, considering its likely manner of installation, to be explosion proof.

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b. Equipment that may come into contact with flammable fluids or vapors, and that under fault conditions may produce arcs, electrical sparks, friction sparks or hot surfaces shall be designed and installed to reduce to an acceptable minimum the overall risk of a fault occurring that will ignite the flammable vapors.

c. In designing the air supply system for forced air ventilated equipment, the possibility of contamination of the air by flammable vapors shall be taken into account. If the equipment and its ducting, including joints, are in an area that can be so contaminated, they shall be capable of meeting the conditions appropriate to the environment.

d. The specification for Category A equipment, paragraph 9.3.1, should consider the design requirements of flange and hole dimensions or other equivalent means, such as flame traps, for adequate safety from flame propagation. This information is contained in national documents.

9.6 Test Procedures

9.6.1 Containment Test

9.6.1.1 Preparation for Test

a. Preparation of Test Case or Enclosure – The purpose of this procedure is to intentionally fill the case or enclosure under test with a volatile gas mixture then ignite it. Drilling and tapping openings in the case or enclosure for inlet/outlet hose connections to the fuel-vapor-air mixture circulation system and for mounting a spark gap device may be required. The case volume shall not be altered by more than ±5 percent by any modification to facilitate the introduction of explosive vapor.

b. Hose Installation - When inserting a hose from a blower, adequate precaution shall be taken to prevent ignition of the ambient mixture by backfire or the release of pressure through the supply hose.

c. Spark Gap Device - A spark gap device for igniting the explosive mixture within the case or enclosure shall be provided. The case or enclosure may be drilled and tapped for the spark gap device, or the spark gap device may be mounted internally.

d. Case Installation - The case or enclosure with either the test item or a model of the test item of the same volume and configuration in position within the case or enclosure shall be connected and oriented in the explosion chamber mechanically and electrically, as recommended by the manufacturer for normal service installation. This shall include any cooling provisions, as necessary to perform the tests described herein.

e. Test Facility – The required apparatus consists of a chamber or cabinet together with auxiliary instrumentation capable of establishing, maintaining and monitoring the specified test conditions. Use a chamber with a means of igniting the fuel-air mixture such as a spark-gap device, as well as a means of determining the explosiveness of a sample of the mixture such as a thermocouple or pressure transducer. An alternative method of determining the explosive characteristics of the vapor is by using a

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calibrated explosive gas meter which verifies the degree of explosiveness and the concentration of the fuel-air mixture.

9.6.1.2 Performance of Test

The following test sequence shall be performed three times as follows:

Step 1: The chamber shall be sealed and the internal pressure maintained at site level pressure. The ambient chamber temperature shall be at least 25°C. An explosive mixture within the chamber shall be obtained by using the mixture defined in paragraphs 9.4.3 and 9.4.4. Circulate (by blower or pump) fuel/air mixture through test unit to insure explosive mixture within test unit.

Step 2: The internal case ignition source shall be energized in order to cause an explosion within the case. The occurrence of an explosion within the case may be detected by use of a thermocouple inserted in the case and connected to an appropriate temperature recorder outside the test chamber. If ignition of the mixture does not occur immediately, the test shall be considered void and shall be repeated with a new explosive charge.

Step 3: At least five internal case explosions shall be performed. If the case tested is small (not in excess of 1/50 of the test chamber volume) and if the reaction within the case upon ignition is of an explosive nature without continued burning of the mixture as it circulates into the case, more than one internal case explosion, but not more than five, may be produced without recharging the entire chamber. If the case volume is greater than 1/50 of the test chamber volume an alternative to purging the chamber after each case explosion would be to verify the explosiveness of the mixture after each case explosion. Ample time shall be allowed between internal case explosions for replacement of burnt gases with fresh explosive mixture within the case. If the internal case explosions produced did not cause a main chamber explosion, the explosiveness of the fuel-air mixture in the main chamber shall be verified by igniting a sample of the mixture with a spark plug or glow plug. If the air-vapor mixture in the main chamber is found not to be explosive, the test shall be considered void and the entire procedure repeated.

9.6.1.3 Failure Criteria

If the internal case explosion causes a main chamber explosion, the test item shall have failed the test and no further tests need be conducted.

9.6.2 Non ignition Test

9.6.2.1 Preparation for Test

a. The test item shall be connected and oriented mechanically and electrically as recommended by the manufacturer for normal service installation. This shall include any cooling provisions as necessary to perform the tests described herein so that

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normal electrical operation is possible and mechanical controls may be operated through the pressure seals from outside the chamber. External covers of the test item shall be removed or loosened to facilitate the penetration of the explosive mixture. Large test items may be tested one or more units at a time by extending electrical connections through the cable port to the balance of the associated equipment located externally.

b. The test item shall be operated to determine that it is functioning properly.

c. Mechanical loads on drive assemblies and servomechanical and electrical loads on switches and relays may be simulated if proper precaution is given to duplicating the normal load in respect to torque, voltage, current, inductive reactance, etc. In all instances, it is preferable to operate the test item as it normally functions in the installed environment.

9.6.2.2 Performance of Test

The test shall be conducted at site level pressure.

Step 1: The test chamber shall be sealed and the ambient temperature within shall be raised to the Operating High Temperature, given in Table 4-1, for which the equipment is designed to operate. The temperature of the test chamber and the chamber walls shall be permitted to rise to within 11°C of the chamber ambient air, prior to the introduction of the explosive mixture, to prevent condensation of the explosive medium.

Step 2: The required quantity of fuel (paragraph 9.4.4) shall be introduced into the chamber. Circulate the test atmosphere for at least three minutes to allow for complete vaporization of fuel and the development of a homogenous mixture.

Step 3: At this time all electrical contacts of the test item shall be actuated. The operation of the test item shall be continuous throughout this period and all making and breaking of electrical contacts shall be conducted as frequently as deemed practicable.

Step 4: If no explosion has occurred as a result of the test item operation, the potential explosiveness of the air-vapor mixture shall be verified by igniting a sample of the mixture with a spark gap or glow plug. If the air-vapor mixture is not found to be explosive, the test shall be considered void and the entire procedure repeated.

9.6.2.3 Failure Criteria

If the item causes an explosion, it shall have failed the test and further tests need not be conducted.

9.6.3 Components or Surface Temperature Test

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9.6.3.1 Preparation for Test

The test item shall be placed in the test chamber in accordance with subparagraph 9.6.2.1. the suspected components or surfaces to be tested for thermal ignition shall be instrumented with thermocouples operating in a range of 65 to 260°C.

9.6.3.2 Performance of Test

The test shall be conducted as follows:

Step 1: The test chamber shall be sealed and the ambient temperature within shall be raised to the Operating High Temperature, given in Table 4-1, for which the equipment is designed to operate.

Step 2: After the equipment stabilizes the equipment shall be turned on and operated in its normal mode until thermal stabilization of the equipment has been attained. The maximum temperatures attained at the suspected components or surfaces shall be recorded continuously from the initial powering of the equipment to its stabilization. If a temperature in excess of 204°C is attained, the test shall be terminated.

9.6.3.3 Failure Criteria

In Step 2 above, if the item exceeds 204°C, the test item shall have failed the test and further tests need not be conducted

Table 9-1 Equipment Categories and Test Requirements

Equipment Categories	Requirements and Tests
A (hermetically sealed)	Containment test 9.6.1 or hermitically sealed case. Surface temperature test 9.6.3.
E	Non ignition test of 9.6.2.
H	Surface temperature test 9.6.3 (non spark producing equipment or hermitically sealed)

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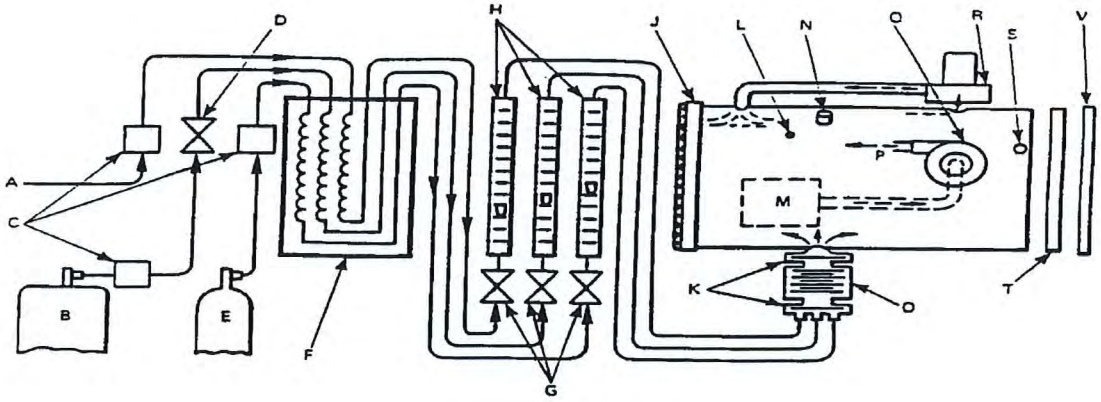

Legend

A AIR
B PROPANE GAS CONTAINER
C REDUCING VALVES
D GAS CUT-OFF VALVE OPERATED BY MICRO-SWITCHES ON EXPLOSION CHAMBER
E OXYGEN BOTTLE
F HEAT EXCHANGER TO BRING THE GASES TO STANDARD TEMPERATURE
G NEEDLE VALVES
J BOTH ENDS COVERED BY DIAPHRAGMS (E G PAPER, POLYETHYLENE) HELD ON BY RUBBER BANDS
K DIAPHRAGM CHECK VALVES
L CYLINDRICAL EXPLOSION CHAMBER
M UNIT UNDER TEST
N VENT
O MIXING CHAMBER
P WASTE TO ATMOSPHERE
Q EXTRACTOR FOR CHARGING UNIT UNDER TEST
R STIRRING BLOWER
S MICRO-SWITCH (ONE T EACH END) RELEASED WHEN H FLOW METERS RUBBER BAND IS DISPLACED BY EXPLOSION
T DIAPHRAGM
V RUBBER BAND

Figure 9-1 Example of Apparatus for Testing in Explosive Atmospheres

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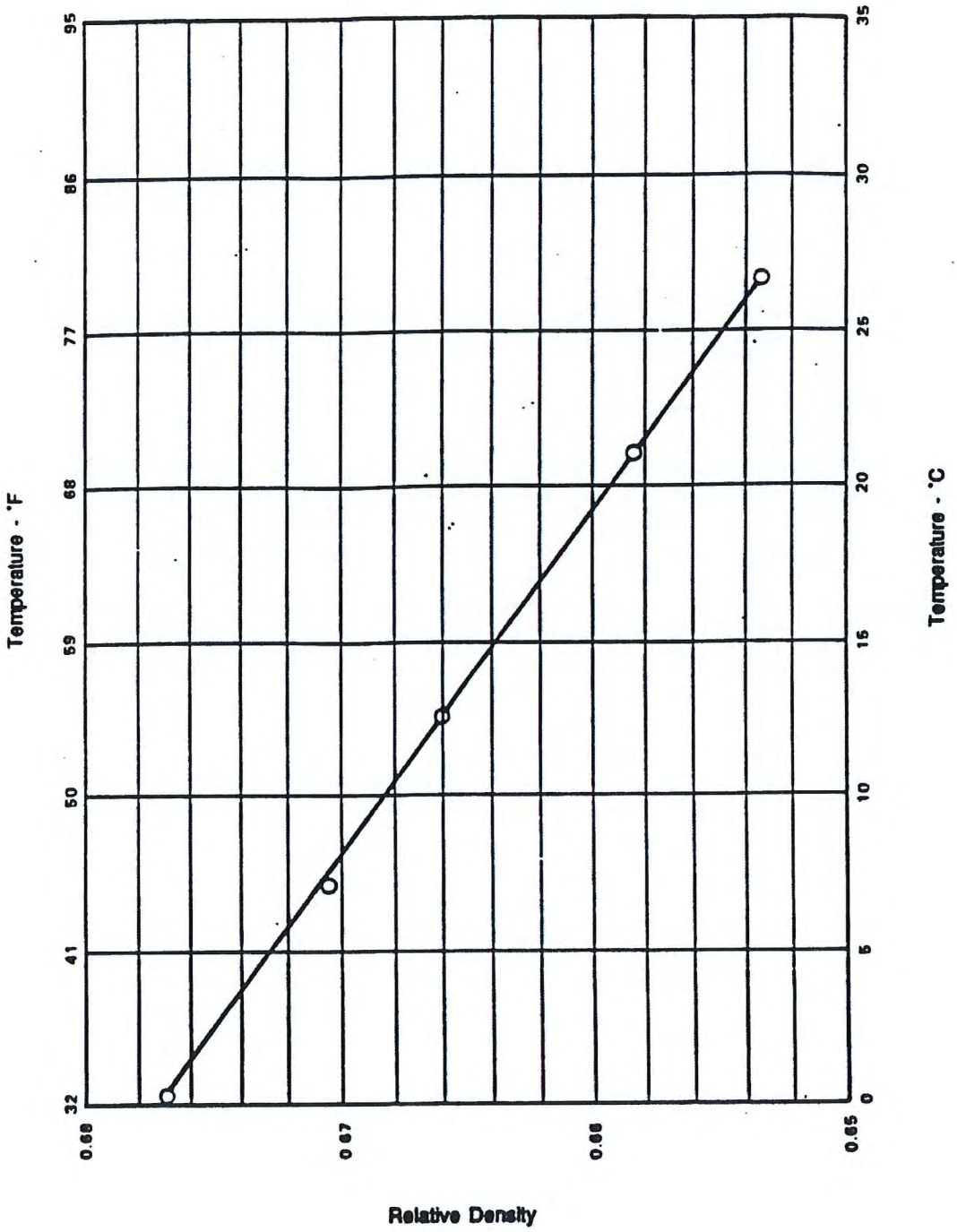

FIGURE 9-2 RELATIVE DENSITY OF N-HEXANE

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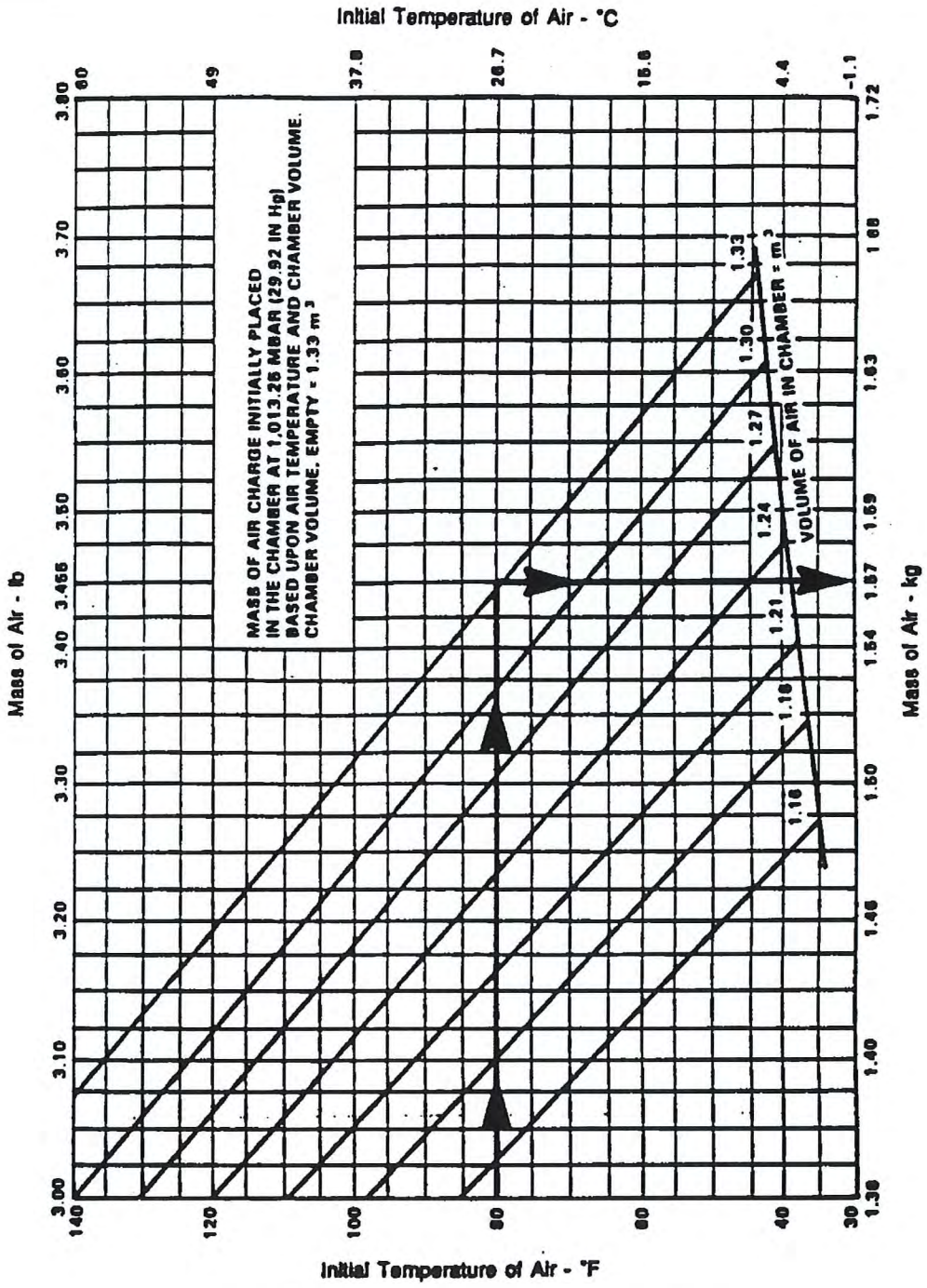

Figure 9-3 Mass of Air Charge vs Temperature

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Environmental Conditions and Test Procedures for Airborne Equipment

User's Guide Appendix 9

Explosive Atmosphere

Based on
DO-160G Section 9.

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User's Guide, Appendix 9

Section 9 defines the environment, test procedures and requirements. In order to properly interpret the procedures and requirements outlined in Section 9, a User's Guide is provided in Appendix 9.

The User's Guide includes rationale, guidance and background information for the environment, test procedures and requirements, guidance in applying the requirements, and lessons learned from aircraft and laboratory experience. This information should help user's understand the intent behind the requirements, should aid in tailoring requirements as necessary for particular applications and should help user's develop detailed test procedures based on the general test procedures in Section 9.

The User's Guide follows a parallel format to the main body of Section 9 for easy cross reference. Paragraph numbers corresponding to each paragraph in Section 9 are included the User's Guide. Where there is no additional information provided in the User's Guide, the paragraph header is provided but the following paragraph is left blank.

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9.A.0 Explosive Atmosphere

9.A.1 Purpose of the Test

This section of DO-160 is intended to provide test procedures and test levels that can be used to test equipment for explosive atmosphere. It is the responsibility of the installer to make sure the test results satisfy the certification requirements of the proposed installation. This user's guide does not contain requirements; it is intended to provide background information and considerations that improve the likelihood of successful test results.

9.A.2 Explosion Proof

9.A.3 Equipment Categories

9.A.3.1 Category A Equipment

9.A.3.2 Category E Equipment

9.A.3.3 Category H Equipment

9.A.4 General Test Requirements

9.A.4.1 General

9.A.4.2 Test Specimen

9.A.4.3 Fuel

9.A.4.4 Fuel Mixture

9.A.5 Equipment Design and Installation Information

Aircraft Zone Definitions and Equipment Requirements

Equipment should be installed in aircraft zones depending on their ability to prevent vapor explosion during their normal functioning and depending on probability that exist in considered zone, during abnormal functioning in failure mode. Definition of Zone provided below.

Aircraft Zone 1

Aircraft Zone 1 is an aircraft area in which uncovered flammable fluids or vapors exist, or can exist, either continuously or intermittently (e.g., in fuel tanks or within fuel or hydraulic systems). Equipment installed in such environment should not be able to ignite the surrounding atmosphere in their normal or double failure conditions of operation.

Aircraft Zone II

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Aircraft Zone II is an aircraft area in which flammable mixtures can be expected to occur only as a result of a single fault-causing spillage or leakage. Equipment installed in such Environment should not be able to ignite the surrounding atmosphere in their normal or faulty possible operation modes. Only equipment single failure should be considered.

Aircraft Zone III

Aircraft Zone III is an aircraft area in which flammable mixtures cannot be expected to occur as a result of a single fault-causing spillage or leakage. Equipment installed in such Environment should not be able to ignite the surrounding atmosphere in their normal operation modes.

Aircraft Zone IV

Aircraft Zone IV is a designated fire zone or an aircraft area in which flammable mixtures cannot be expected to occur as a result of a double fault-causing spillage or leakage. No requirement applies to equipment located in such zone.

9.A.6 Test Procedures

9.A.6.1 Containment Test

9.A.6.1.1 Preparation for Test

9.A.6.1.2 Performance of Test

9.A.6.1.3 Failure Criteria

9.A.6.2 Non Ignition Test

9.A.6.2.1 Preparation of Test

9.A.6.2.2 Performance of Test

9.A.6.2.3 Failure Criteria

9.A.6.3 Components of Surface Temperature Test

9.A.6.3.1 Preparation for Test

9.A.6.3.2 Performance of Test

9.A.6.3.3 Failure Criteria

If faulty modes are to be tested, the equipment should be turned on and operated in its possible faulty modes until thermal stabilization of the equipment has been attained. The temperatures attained at the suspected components or surfaces should be recorded continuously from the initial powering of the equipment to its stabilization. If a temperature in excess of 204°C is attained at any time during the period, the test should be terminated.

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Table 9-A-1 **Equipment Categories and Test Requirements**

Figure 9-A-1 **Example of Apparatus for Testing in Explosive Atmospheres**

Figure 9-A-2 **Relative Density of N-Hexane**

Figure 9-A-3 **Mass of Air Charge vs. Temperature**

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